PURE COMPONENT THERMODYNAMIC PROPERTY PREDICTOR

Software Project Report

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Abstract

This project presents an interactive software application for estimating thermodynamic properties of pure real gases. It implements the Peng–Robinson (PR) and Soave–Redlich–Kwong (SRK) cubic equations of state and calculates PVT behaviour, compressibility factor, fugacity coefficient, fugacity, and enthalpy and entropy changes. A Streamlit interface allows users to select a component, equation of state, phase-root option, and operating conditions, and inspect calculated values and plots. The application is intended as an educational engineering tool. Its constant heat-capacity approximation and the absence of independent experimental validation should be considered when interpreting results.

1. Introduction and Objectives

Real gases deviate from ideal-gas behaviour because of intermolecular attractions and finite molecular volume. Cubic equations of state provide a practical framework for representing these deviations and estimating volumetric and related thermodynamic properties. This project translates two commonly used cubic models into an interactive computational tool for pure-component calculations.

The objectives are to predict PVT behaviour using PR and SRK; calculate enthalpy and entropy changes by combining ideal-gas and residual contributions; calculate fugacity coefficient and fugacity; compare the two equations of state; and provide plots and an interface for exploring operating conditions and component properties.

2. Thermodynamic Model

The software uses critical temperature (Tc), critical pressure (Pc), and acentric factor (ω) to calculate the temperature-dependent attraction parameter a(T) and co-volume parameter b. The alpha function is $\alpha = [1 + m(1 - \sqrt{(T/Tc)})]^2$, with m determined from the acentric factor using the correlation implemented for the selected EOS.

For Peng–Robinson, $ac = 0.45724\ R^2Tc^2/Pc$ and $b = 0.07780\ RTc/Pc$. For Soave–Redlich–Kwong, $ac = 0.42747\ R^2Tc^2/Pc$ and $b = 0.08664\ RTc/Pc$. Dimensionless parameters A and B are used to form the corresponding cubic equation in compressibility factor Z. Physically admissible real roots are retained for phase selection.

Molar volume is calculated from $V = ZRT/P$. Fugacity is evaluated using $f = \varphi P$, where φ is the fugacity coefficient obtained from the selected EOS expression. The software also reports ln φ.

3. Enthalpy and Entropy Calculations

Enthalpy and entropy changes combine ideal-gas changes with EOS departure (residual) properties. The ideal-gas contributions use a constant component heat capacity Cp: $\Delta Hig = Cp(T - Tref)$ and $\Delta Sig = Cp\ \ln(T/Tref) - R\ \ln(P/Pref)$. Residual contributions are calculated from cubic EOS parameters, the selected compressibility root, and the temperature derivative of the attraction parameter.

The reported totals are $\Delta H = \Delta Hig + Hres$ and $\Delta S = \Delta Sig + Sres$. Reference temperature and pressure are user-selectable in the H/S interface. These outputs are property changes relative to the selected reference state, not absolute enthalpy or entropy values.

4. Software Design and Implementation

The application is written in Python and uses Streamlit for the interactive interface, NumPy for numerical operations, Pandas for data handling, and Matplotlib for plotting. The built-in component database contains methane, ethane, propane, n-butane, nitrogen, carbon dioxide, oxygen, hydrogen, water, and ammonia. A custom-component input option allows users to enter component properties.

The interface contains four main areas: PVT, H/S, Fugacity, and Model & Assumptions. PVT displays state properties and pressure–volume and compressibility-factor plots. H/S presents ideal, residual, and total enthalpy and entropy changes. Fugacity displays φ, ln φ, and f. The model information area describes equations, units, assumptions, and limitations.

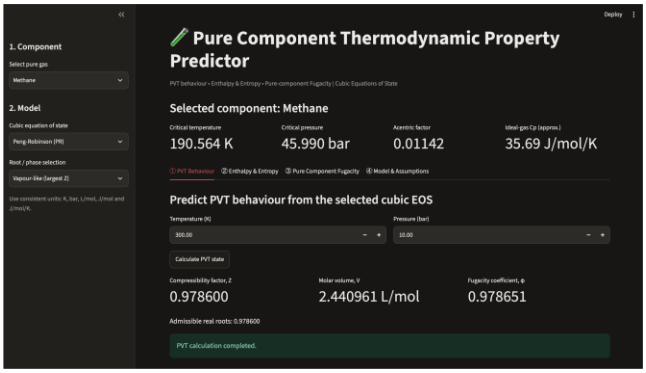
5. Demonstration Results

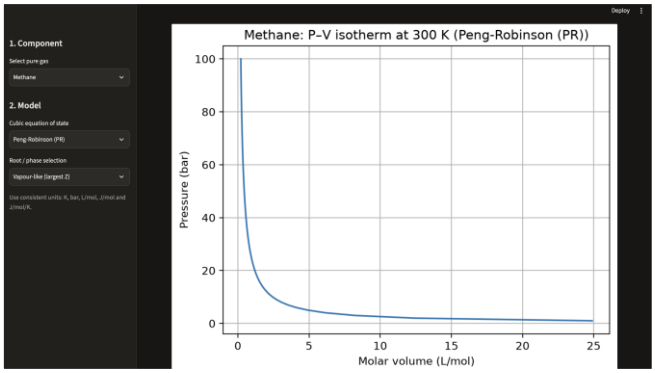
A demonstration calculation was performed for methane at 300 K and 10 bar using the built-in methane properties. For the H/S illustration, the reference state was 298.15 K and 1 bar. The values below are application outputs; they demonstrate calculation and comparison features and are not claimed as experimental validation.

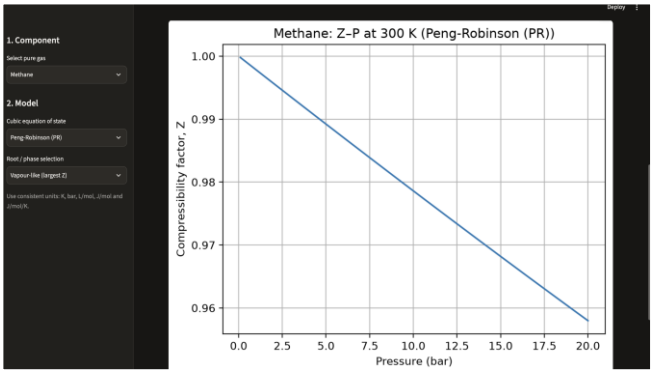
Demonstration data: methane at 300 K and 10 bar

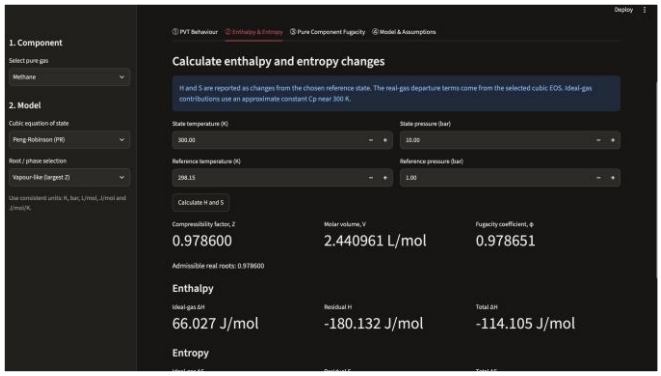
Quantity	PR EOS	SRK EOS	Unit
Compressibility factor, Z	0.978600	0.983594	—
Molar volume, V	2.440961	2.453417	L/mol
Fugacity coefficient, φ	0.978651	0.983617	—
Fugacity, f	9.786508	9.836168	bar
Total enthalpy change, ΔH	−114.105	−98.544	J/mol
Total entropy change, ΔS	−19.345	−19.335	J/(mol·K)

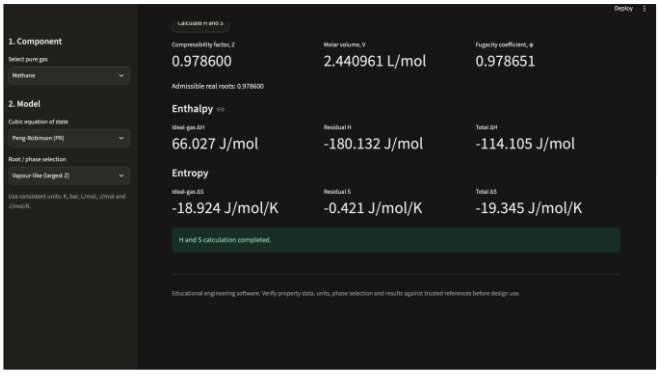
H/S reference state: 298.15 K and 1 bar. Values are rounded

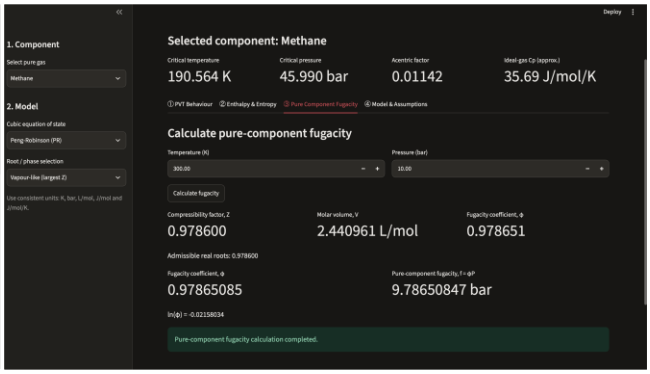


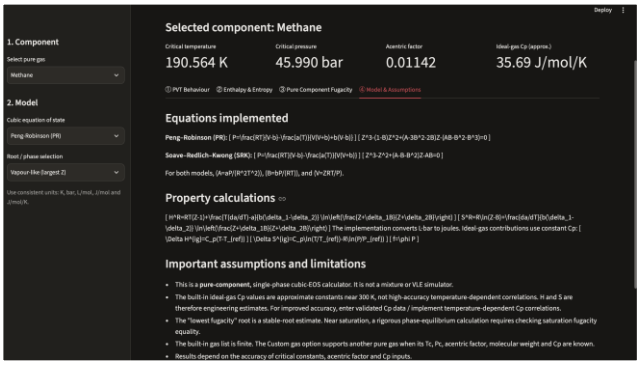


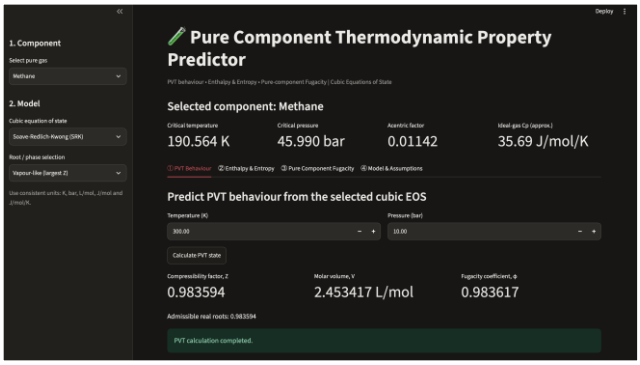


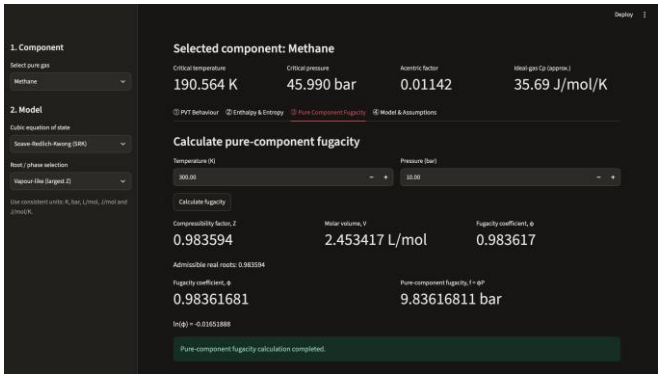


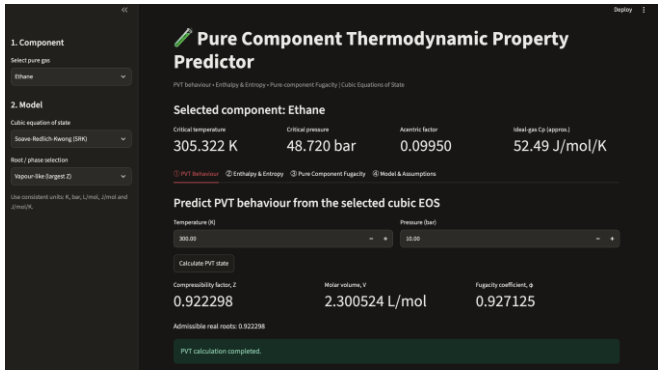


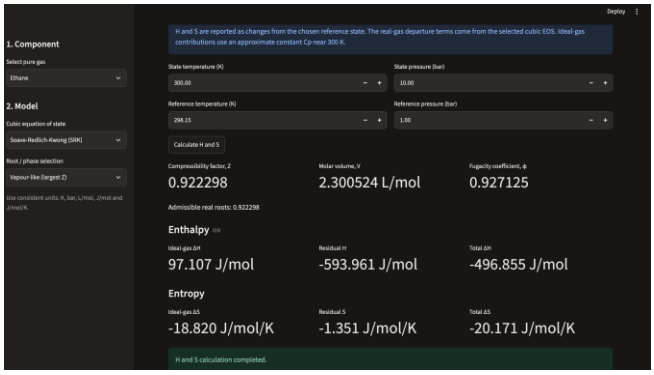



Deploy
1. Component
Select pure gas
Methane
2. Model
Cubic equation of state
Peng-Robinson (PR)
Root / phase selection
Vapour-like (largest Z)
Use consistent units: K, bar, L/mol, J/mol and J/mol/K.
Selected component: Methane
Critical temperature
190.564 K
Critical pressure
45.990 bar
Acentric factor
0.01142
Ideal-gas Cp (approx.)
35.69 J/mol/K
① PVT Behaviour ② Enthalpy & Entropy ③ Pure Component Fugacity ④ Model & Assumptions
Equations implemented
Peng–Robinson (PR): [P=\frac{RT}{V-b}-\frac{a(T)}{V(V+b)+b(V-b)}] [Z^3-(1-B)Z^2+(A-3B^2-2B)Z-(AB-B^2-B^3)=0]
Soave–Redlich–Kwong (SRK): [P=\frac{RT}{V-b}-\frac{a(T)}{V(V+b)}] [Z^3-Z^2+(A-B-B^2)Z-AB=0]
For both models, (A=aP/(R^2T^2)), (B=bP/(RT)), and (V=ZRT/P).
Property calculations
[H^R=RT(Z-1)+\frac{T\da/dT}-a}{b\delta_1-\delta_2}\ln\left(\frac{Z+\delta_1B}{Z+\delta_2B}\right)] [S^R=R\ln(Z-B)+\frac{da/dT}{b\delta_1-\delta_2}\ln\left(\frac{Z+\delta_1B}{Z+\delta_2B}\right)] The implementation converts L·bar to joules. Ideal-gas contributions use constant Cp: [\Delta H^{ig}=C_p(T-T_{ref})] [\Delta S^{ig}=C_p\ln(T/T_{ref})-R\ln(P/P_{ref})] [f=\phi P]
Important assumptions and limitations
This is a pure-component, single-phase cubic-EOS calculator. It is not a mixture or VLE simulator.
The built-in ideal-gas Cp values are approximate constants near 300 K, not high-accuracy temperature-dependent correlations. H and S are therefore engineering estimates. For improved accuracy, enter validated Cp data / implement temperature-dependent Cp correlations.
The "lowest fugacity" root is a stable-root estimate. Near saturation, a rigorous phase-equilibrium calculation requires checking saturation fugacity equality.
The built-in gas list is finite. The Custom gas option supports another pure gas when its Tc, Pc, acentric factor, molecular weight and Cp are known.
Results depend on the accuracy of critical constants, acentric factor and Cp inputs.


Deploy
1. Component
Select pure gas
Methane
2. Model
Cubic equation of state
Soave-Redlich-Kwong (SRK)
Root / phase selection
Vapour-like (largest Z)
Use consistent units: K, bar, L/mol, J/mol and J/mol/K.
Pure Component Thermodynamic Property Predictor
PVT behaviour • Enthalpy & Entropy • Pure-component Fugacity | Cubic Equations of State
Selected component: Methane
Critical temperature
190.564 K
Critical pressure
45.990 bar
Acentric factor
0.01142
Ideal-gas Cp (approx.)
35.69 J/mol/K
① PVT Behaviour ② Enthalpy & Entropy ③ Pure Component Fugacity ④ Model & Assumptions
Predict PVT behaviour from the selected cubic EOS
Temperature (K)
300.00
Pressure (bar)
10.00
Calculate PVT state
Compressibility factor, Z
0.983594
Molar volume, V
2.453417 L/mol
Fugacity coefficient, ϕ
0.983617
Admissible real roots: 0.983594
PVT calculation completed.


Deploy
1. Component
Select pure gas
Methane
2. Model
Cubic equation of state
Soave-Redlich-Kwong (SRK)
Root / phase selection
Vapour-like (largest Z)
Use consistent units: K, bar, L/mol, J/mol and J/mol/K.
Selected component: Methane
Critical temperature
190.564 K
Critical pressure
45.990 bar
Acentric factor
0.01142
Ideal-gas Cp (approx.)
35.69 J/mol/K
① PVT Behaviour ② Enthalpy & Entropy ③ Pure Component Fugacity ④ Model & Assumptions
Calculate pure-component fugacity
Temperature (K)
300.00
Pressure (bar)
10.00
Calculate fugacity
Compressibility factor, Z
0.983594
Molar volume, V
2.453417 L/mol
Fugacity coefficient, ϕ
0.983617
Admissible real roots: 0.983594
Fugacity coefficient, ϕ
0.98361681
Pure-component fugacity, f = ϕP
9.83616811 bar
ln(ϕ) = -0.01651888
Pure-component fugacity calculation completed.


Deploy
1. Component
Select pure gas
Ethane
2. Model
Cubic equation of state
Soave-Redlich-Kwong (SRK)
Root / phase selection
Vapour-like (largest Z)
Use consistent units: K, bar, L/mol, J/mol and J/mol/K.
Pure Component Thermodynamic Property Predictor
PVT behaviour • Enthalpy & Entropy • Pure-component Fugacity | Cubic Equations of State
Selected component: Ethane
Critical temperature
305.322 K
Critical pressure
48.720 bar
Acentric factor
0.09950
Ideal-gas Cp (approx.)
52.49 J/mol/K
① PVT Behaviour ② Enthalpy & Entropy ③ Pure Component Fugacity ④ Model & Assumptions
Predict PVT behaviour from the selected cubic EOS
Temperature (K)
300.00
Pressure (bar)
10.00
Calculate PVT state
Compressibility factor, Z
0.922298
Molar volume, V
2.300524 L/mol
Fugacity coefficient, ϕ
0.927125
Admissible real roots: 0.922298
PVT calculation completed.


Deploy
1. Component
Select pure gas
Ethane
2. Model
Cubic equation of state
Soave-Redlich-Kwong (SRK)
Root / phase selection
Vapour-like (largest Z)
Use consistent units: K, bar, L/mol, J/mol and J/mol/K.
H and S are reported as changes from the chosen reference state. The real-gas departure terms come from the selected cubic EOS. Ideal-gas contributions use an approximate constant Cp near 300 K.
State temperature (K)
300.00
State pressure (bar)
10.00
Reference temperature (K)
298.15
Reference pressure (bar)
1.00
Calculate H and S
Compressibility factor, Z
0.922298
Molar volume, V
2.300524 L/mol
Fugacity coefficient, ϕ
0.927125
Admissible real roots: 0.922298
Enthalpy
Ideal-gas ΔH
97.107 J/mol
Residual H
-593.961 J/mol
Total ΔH
-496.855 J/mol
Entropy
Ideal-gas ΔS
-18.820 J/mol/K
Residual S
-1.351 J/mol/K
Total ΔS
-20.171 J/mol/K
H and S calculation completed.

6. Verification and Testing

The source code was checked for Python compilation using python3 -m compileall -q app.py tests/test_core.py. The automated pytest suite completed with 14 passed and 0 failed in 0.63 seconds on Python 3.14.2 (macOS), using pytest 9.2.0. Tests cover methane states with both EOS models, fugacity relationships, ideal-gas limiting behaviour, enthalpy and entropy calculations, residual properties, invalid temperature and pressure handling, cubic-root sorting, and H/S reference-state calculations. The application was also launched and its main interface, tabs, and plots were manually checked.

These tests assess internal consistency, selected limiting behaviour, and input validation. They do not constitute independent comparison against experimental measurements or an external reference-property database.

7. Assumptions and Limitations

The current implementation is limited to pure components. Built-in ideal-gas heat capacities are approximate constant values near 300 K, rather than temperature-dependent Cp correlations. This approximation can reduce accuracy over broad temperature intervals. Results also depend on the quality of component critical properties and acentric factor. Root selection and phase interpretation require care near saturation and critical conditions. The software should be used for learning and preliminary estimation, and predictions should be checked against trusted thermodynamic references before engineering design or safety-critical use.

8. Conclusion

The project delivers an interactive Python application that applies PR and SRK cubic equations of state to pure-component PVT, fugacity, enthalpy-change, and entropy-change calculations. It combines numerical routines, visualisation, a component database, and automated tests in a single interface. The current test suite passes all 14 checks. Further development could include temperature-dependent ideal-gas heat-capacity correlations, independent validation against reference data, additional component data, and mixture calculations.

Execution: python3 -m pip install -r requirements.txt; python3 -m streamlit run app.py

Testing: python3 -m pytest tests/test_core.py -v. See TEST_RESULTS.txt.